

Wishlist Context Wireframes

Overview

Wishlist Context adds semantic labels to wishlisted elements, helping collectors prioritize purchases based on intent. Four context types capture different collecting motivations: Core (foundation-building), Exploration (curiosity-driven), Aspirational (rare/valuable targets), and Upgrade Target (replacement/improvement).

Context Types

1. Core

Definition: Essential elements for foundation-building

Typical Use: Common, affordable elements needed to reach key milestones

Auto-Detection Criteria:

- Element is in a category where user has <50% completion
- Element is priced <\$50
- Element has >70% global ownership (common element)

Examples: Carbon, Copper, Iron, Aluminum

2. Exploration

Definition: Curiosity-driven additions

Typical Use: Interesting but not critical elements

Auto-Detection Criteria:

- Element is in a category where user has 50-90% completion
- Element is priced \$20-\$100
- Element has 30-70% global ownership

Examples: Bismuth, Gallium, Selenium, Tellurium

3. Aspirational

Definition: Rare, valuable, or difficult-to-acquire elements

Typical Use: Long-term goals, “dream” elements

Auto-Detection Criteria:

- Element is priced >\$100
- Element has <20% global ownership (rare)
- Element is in Precious Metals or Rare Earth categories

Examples: Rhodium, Osmium, Iridium, Gold

4. Upgrade Target

Definition: Replacement or improvement of already-owned element

Typical Use: Larger format, better condition, different form

Auto-Detection Criteria:

- User already owns this element in a different format
- New format is larger (e.g., user owns 10mm, wants 50mm)
- New format is more valuable (e.g., user owns ampoule, wants Lucite cube)

Examples: User owns Iron 10mm, wants Iron 50mm cube

UI Wireframes

A. Enhanced Wishlist View

Route: `/app/wishlist`

My Wishlist

[Add Element ▼]

🔥 23 elements on your wishlist

Filter by Context: [All] [Core] [Exploration] [Aspirational] [Upgrade Target]

CORE (8 elements)

[Collapse ▲]

Foundation-building essentials

Zinc (Zn)

🟢 Core

\$28.99

✅ In Stock

[Edit Label]

[View]

Silicon (Si)

🟢 Core

\$32.50

✅ In Stock

[Edit Label]

[View]

Sulfur (S)

🟢 Core

\$19.99

⚠️ Low Stock

[Edit Label]

[View]

Nickel (Ni)

🟢 Core

\$38.00

✅ In Stock

[Edit Label]

[View]

[+4 more...]

EXPLORATION (7 elements)

[Collapse ▲]

Curiosity-driven additions

Gallium (Ga)

🟡 Explore

\$49.99

⚠️ Out Stock

[Edit Label]

[Notify Me]

Bismuth (Bi)

🟡 Explore

\$52.50

✅ In Stock

[Edit Label]

[View]

Selenium (Se)

🟡 Explore

\$44.99

⚠️ Out Stock

[Edit Label]

[Notify Me]

[+4 more...]

ASPIRATIONAL (5 elements)

[Collapse ▲]

Long-term goals and rare finds

Rhodium (Rh)

🟡 Aspire

\$847.00

⚠️ Out Stock

Osmium (Os)

🟡 Aspire

\$1,249.99

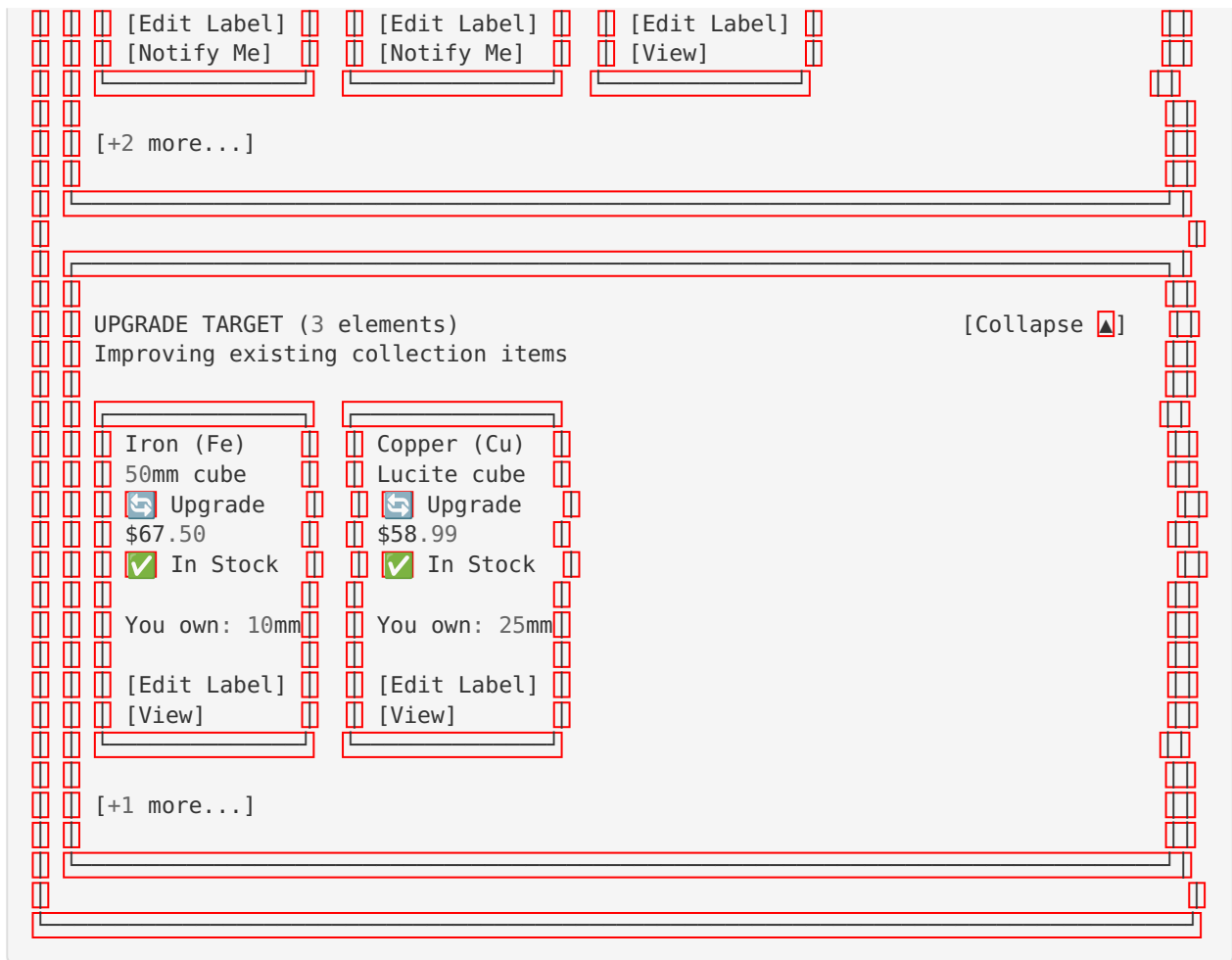
⚠️ Out Stock

Iridium (Ir)

🟡 Aspire

\$923.50

✅ In Stock



Key Features:

- Wishlist items grouped by context type
- Collapsible sections
- Visual badges with distinct colors per context
- Stock status indicators
- “Edit Label” button for manual override
- “Upgrade Target” shows what user already owns

B. Context Label Assignment Modal

Trigger: Click “Edit Label” on any wishlist item

Assign Context **for** Gallium (Ga) [X Close]

Why **do** you want this element?

☐ Core
Foundation-building essentials
☐ Common elements needed **for** milestones

☒ Exploration (Recommended)
Curiosity-driven additions
☐ Interesting but not critical

☐ Aspirational
Long-term goals and rare finds
☐ High-value or hard-to-acquire elements

☐ Upgrade Target
Improving existing items
☐ You don't own Ga yet, so this doesn't fit (grayed **out**, cannot select)

💡 We recommended "**Exploration**" based on:

- You're 68% complete **in** Post-Transition Metals
- Gallium is priced at \$49.99 (mid-range)
- 42% of collectors own this element

[Cancel] [Save Label]

Key Features:

- Radio button selection (single choice)
- Recommended option highlighted
- Explanation for recommendation
- Invalid options grayed out with explanation
- Descriptive subtitles for each context type

C. Auto-Detection Prompt (First-Time Wishlist Add)

Trigger: User adds first element to wishlist (or first 5 elements)



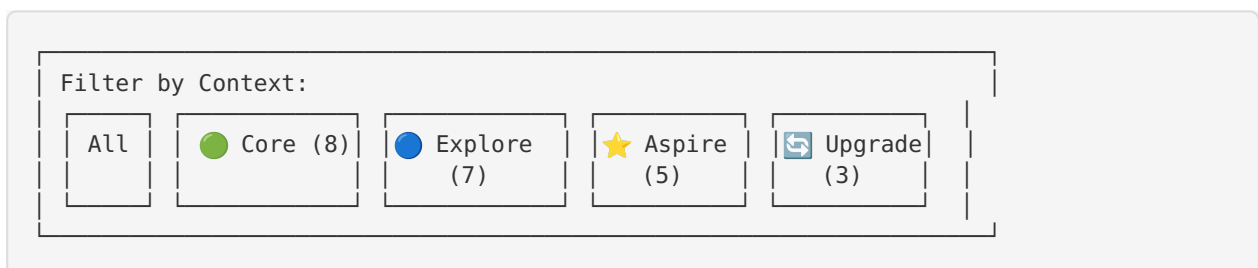
Options:

- **Skip:** Don't assign a label, just add to wishlist
- **Use Core:** Accept the recommendation
- **Choose Different:** Opens full context selection modal

Auto-assign checkbox: If checked, future wishlist additions get auto-labeled (with option to edit)

D. Filter by Context (Compact View)

Filter Bar (at top of wishlist):



Interaction:

- Click a context chip → filters wishlist to show only that context
- Shows count in parentheses
- "All" chip selected by default
- Active chip has blue border/background

Filtered View (e.g., Core selected):

Filter by Context:

All

Core (8)

Explore

Aspire

Upgrade

Active

Showing 8 Core elements

Zinc (Zn)

Core

\$28.99

[View]

Silicon (Si)

Core

\$32.50

[View]

Sulfur (S)

Core

\$19.99

[View]

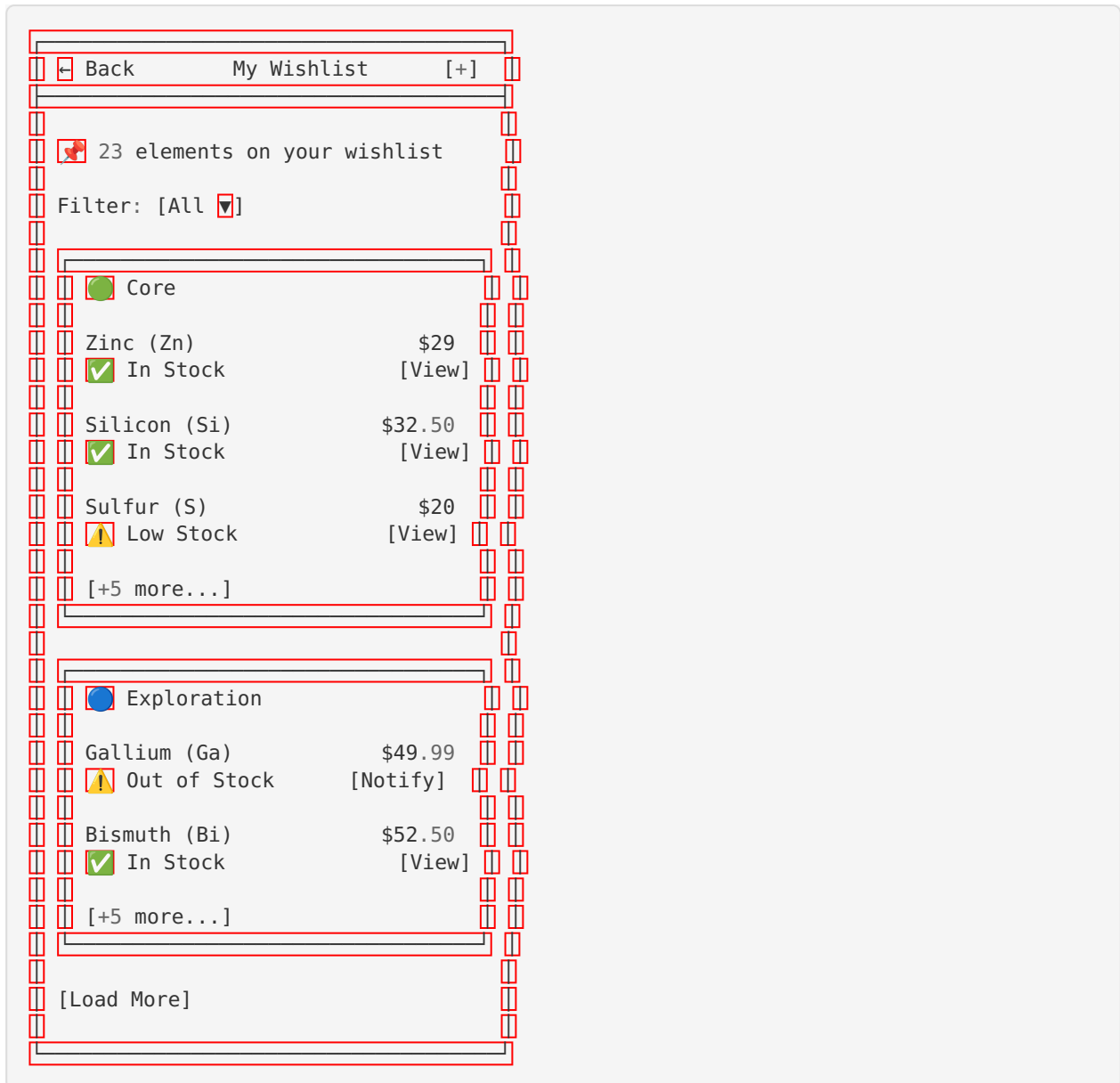
Nickel (Ni)

Core

\$38.00

[View]

E. Mobile Wishlist with Context Labels



Mobile Optimizations:

- Simplified card layout (no photos, just text)
- Larger touch targets
- Dropdown filter instead of chip buttons
- Collapsible sections by default

Auto-Detection Logic

Detection Flow

```

User adds Element X to wishlist
├── Get user's collection stats
│   ├── • Total completion %
│   ├── • Category completion %
│   └── • Owned formats for this element
├── Get element metadata
│   ├── • Price
│   ├── • Global ownership %
│   ├── • Category
│   └── • Rarity score
├── Run context detection algorithm
├── Determine context:
│   ├── IF user owns element in different format
│   │   └── Context = UPGRADE_TARGET ✓
│   ├── ELSE IF price > $100 OR global ownership < 20%
│   │   └── Context = ASPIRATIONAL ✓
│   ├── ELSE IF category completion 50-90%
│   │   └── Context = EXPLORATION ✓
│   └── ELSE (category completion < 50% AND common)
│       └── Context = CORE ✓
├── Show context label on wishlist item
└── Store context in WishlistItem.contextLabel field
  
```

Algorithm Pseudocode

```
function detectWishlistContext(userId, elementSymbol, productId) {
  // Check if user owns this element in a different format
  const ownedFormats = await getUserOwnedFormats(userId, elementSymbol);
  const thisProductFormat = await getProductFormat(productId);

  if (ownedFormats.length > 0 && !ownedFormats.includes(thisProductFormat)) {
    return {
      context: 'UPGRADE_TARGET',
      reason: `You own ${elementSymbol} in ${ownedFormats[0]} format`
    };
  }

  // Get element and product metadata
  const product = await getProduct(productId);
  const element = await getElement(elementSymbol);
  const globalOwnership = await getGlobalOwnershipPercent(elementSymbol);

  // Check if aspirational (rare or expensive)
  if (product.price > 100 || globalOwnership < 20 || element.category === 'Precious Metals') {
    return {
      context: 'ASPIRATIONAL',
      reason: `High value or rare element (${globalOwnership}% ownership)`
    };
  }

  // Check category completion
  const categoryCompletion = await getCategoryCompletion(userId, element.category);

  // If category is well-developed, it's exploration
  if (categoryCompletion >= 50 && categoryCompletion < 90) {
    return {
      context: 'EXPLORATION',
      reason: `You're ${categoryCompletion}% complete in ${element.category}`
    };
  }

  // Default: Core (foundation-building)
  return {
    context: 'CORE',
    reason: `Common element for building your ${element.category} collection`
  };
}
```

Context Override Flow

User Manual Override

```
User clicks "Edit Label" on Zinc (currently labeled "Core")
├─ Context selection modal opens
│   ├── Core (currently selected)
│   ├── Exploration
│   ├── Aspirational
│   └── Upgrade Target (grayed out - user doesn't own Zn yet)
├─ User selects "Aspirational"
├─ System updates WishListItem.contextLabel = 'ASPIRATIONAL'
├─ System sets WishListItem.contextOverridden = true
├─ Zinc now appears in "Aspirational" section
└─ Future auto-detection won't change this label (respects user override)
```

Key Point: Once user manually overrides, system never auto-changes it again

Database Schema

Updated WishListItem Model

```
model CollectionItem {
  id                String                @id @default(cuid())
  userId            String
  elementSymbol     String
  state             CollectionItemState // 'owned' | 'wishlist' | 'watchlist' | 'missing'

  // NEW FIELDS FOR CONTEXT
  contextLabel      WishlistContext?
  contextOverridden Boolean              @default(false)
  contextReason     String?              // Auto-detection explanation

  priority          Int?                 @default(0)
  createdAt         DateTime              @default(now())
  updatedAt         DateTime              @updatedAt

  user              User                  @relation(fields: [userId], references: [id])
  element           Element               @relation(fields: [elementSymbol],
  references: [symbol])
}

enum WishlistContext {
  CORE
  EXPLORATION
  ASPIRATIONAL
  UPGRADE_TARGET
}
```

Field Descriptions:

- contextLabel : The assigned context (null if none)
 - contextOverridden : True if user manually changed it (prevents auto-updates)
 - contextReason : Explanation text shown to user (e.g., "You're 68% complete in Post-Transition Metals")
-

Admin View Enhancement

Admin Demand Insights: Context Breakdown

Location: /app/admin/demand (existing route)

New Section: Context Distribution

Wishlist Context Distribution

Understanding collector intent across all wishlists

Context Breakdown (All Collectors)

Context	Total Items	Avg per User	%
 Core	1,876	7.6	42.3%
 Exploration	1,542	6.2	34.7%
 Aspirational	789	3.2	17.8%
 Upgrade	234	0.9	5.2%

Most Wanted by Context

CORE Elements (Top 5)

Element	Wishlist Cnt	In Stock?	Price
Zinc (Zn)	187	 Yes (45)	\$28.99
Silicon (Si)	176	 Yes (32)	\$32.50
Sulfur (S)	164	 Low (8)	\$19.99
Nickel (Ni)	152	 Yes (28)	\$38.00
Titanium (Ti)	147	 Yes (51)	\$45.50

ASPIRATIONAL Elements (Top 5)

Element	Wishlist Cnt	In Stock?	Price
Rhodium (Rh)	89	 No	\$847
Osmium (Os)	76	 No	\$1,250
Iridium (Ir)	68	 Yes (2)	\$924
Platinum (Pt)	63	 Yes (5)	\$672
Palladium (Pd)	58	 No	\$589

[Export Context Analysis]

Insights for Staff:

- See which context types are most common
- Identify high-demand items per context (helps prioritize restocking)
- Understand collector intent (e.g., if 42% are "Core", focus on affordable essentials)

User Benefits

1. Prioritization

- Quick glance at wishlist shows what's essential vs. aspirational
- Filter by context to focus shopping budget

2. Budget Planning

- "Core" items are affordable foundation-builders
- "Aspirational" items are saved for special occasions
- "Exploration" items are flexible curiosity purchases

3. Collection Strategy

- See at a glance how many "Core" items are left to reach milestone
- Track upgrade targets separately
- Celebrate when aspirational items come in stock

Success Metrics

For Users

- **Wishlist clarity:** Users can articulate why each element is on wishlist
- **Purchase decisions:** Faster decision-making when shopping
- **Engagement:** More frequent wishlist check-ins

For Business

- **Conversion rate:** Higher conversion on in-stock wishlist items
- **Average order value:** Core + Exploration bundling
- **Restock prioritization:** Focus on high-demand context types

Future Enhancements (Post-MVP)

Phase 3+ Features

1. **Smart Sort:** Sort wishlist by context + stock status + price
 2. **Budget Planner:** "Shop Core items for <\$200"
 3. **Bundle Suggestions:** "Complete your Core collection - 5 items for \$120"
 4. **Notification Priority:** Notify sooner for Core restocks, later for Aspirational
 5. **Goal Tracking:** "3 more Core items to reach 50% completion"
-

Design System

Context Color Palette

Context	Color	Badge	Emoji
Core	#48BB78 (green)		
Exploration	#4299E1 (blue)		
Aspirational	#F6AD55 (gold)		
Upgrade Target	#9F7AEA (purple)		

Typography

- **Context Label:** 12px, bold, uppercase, colored by context
 - **Context Description:** 14px, gray, regular weight
-

Summary

Purpose: Add semantic meaning to wishlist items

Context Types: 4 (Core, Exploration, Aspirational, Upgrade Target)

Auto-Detection: Smart labeling based on collection state + price + rarity

Override: User can manually change any label

Benefits: Better prioritization, budget planning, collection strategy

Philosophy: Not all wishlist items are equal. Context helps collectors focus their budget and energy on what matters most right now.